From pairwise electrocoalescence to optical clearing: a predictive dimensionless framework for electrostatic demulsification of nanoemulsions

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Abstract. The pair-scale physics of electrocoalescence is textbook material: water droplets in oil polarize, attract, chain, and merge. Yet the observable that decides whether an electrostatic demulsifier works, the time for the emulsion to *optically clear*, cannot be obtained from any single-pair argument, and has so far been measured rather than predicted. Here we construct a predictive chain from pair electrostatics to the operational 95%-transmittance endpoint for water-in-oil nanoemulsions, using only pre-experiment setup information: material properties, cell geometry, device characterization, the initial droplet-size distribution, and a bound on the surfactant film barrier extracted from the emulsion's zero-field shelf life. Three independent computational routes, a closed-form staged cascade, a polydisperse population balance with a Mie-scattering endpoint, and a stochastic super-droplet simulation steered by a reinforcement-learning agent, predict, without calibration to any kinetic measurement, clearing in 90–125 minutes for a 175 nm, 2% emulsion at 7.5 kV/cm; the measured value, not consulted during model construction, is ∼60 minutes. The framework identifies two well-separated clocks: bulk water separation (minutes) and optical clearing (hours), the latter controlled by a sub-100-nm fines tail. A single dimensionless group, the electrocoalescence number $\Lambda = \pi\varepsilon_0\varepsilon_{\mathrm{oil}}K^2E^2a^3/k_BT$, gates the entire cascade: published voltage-sweep data collapse onto $t \cdot \Lambda = 23.5$ min (5% RMS scatter), with the go/no-go threshold $\Lambda^* \approx 0.8$ predicted in advance. The framework further explains, without fitting, why saline droplets demulsify more slowly than fresh water. Code and data are openly available.

1 Introduction

Separating emulsified water from oil is a rate-limiting operation across crude desalting, produced-water treatment, and waste-oil recycling [21, 22], and it becomes hardest exactly where it matters most: at nanoscale droplet sizes, where gravity is powerless and surfactant films render the dispersion metastable for months [1, 2, 23, 24]. Electrocoalescence, driving droplet merger with an electric field, is the classical remedy [3, 1, 25, 26], but conventional immersed-electrode devices are capped near 1 kV/cm by shorting through droplet chains. A recently demonstrated non-Laplacian, air-gap corona architecture removes that cap and sustains fields up to ∼7.5 kV/cm across the emulsion, demulsifying 150–800 nm water-in-oil emulsions without chemical additives [4].

The microphysics underlying such devices is well established. A conducting aqueous droplet in oil acquires an induced dipole; aligned dipoles attract and chain, a fibration known since Winslow's early electrorheology [27, 28], and press out the intervening oil film until rupture and merger follow [3, 5, 6, 29]. What is *not* established is the bridge from this pair-scale picture to the quantity an operator actually observes: the time for a milky emulsion to become transparent. The two are separated by seven orders of magnitude in droplet count, by a coalescence cascade spanning three decades in radius, by gravity, and by the optics of light scattering. Population-balance models of electrocoalescers have made important progress on parts of this bridge, predicting droplet-size evolution and water-removal efficiency in microscale systems [16, 17, 18, 19, 20], but they are calibrated against measured kinetics, address micron-scale emulsions, and stop short of the observable an operator actually uses: the optical clearing of the sample. For the nanoemulsion regime, where Brownian motion, surfactant film barriers, and light scattering by sub-wavelength droplets dominate, the clearing time has remained an empirical quantity: measured and correlated with voltage, but not predicted from first principles.

This paper constructs and tests that bridge. Our contributions are fourfold. First, a multiscale framework that converts setup information alone, material properties, geometry, device current–voltage characterization, the initial size distribution, and a film-barrier bound extracted from the emulsion's zero-field shelf life, into a prediction of the *optical* clearing time with zero calibrated parameters; unlike prior population-balance treatments [16, 17, 18], nothing is fitted to kinetic data, the endpoint is evaluated through Mie optics exactly as measured, and the regime is the Brownian-dominated nanoemulsion limit. Second, a strict separation of two observables that are usually conflated: the *mass clock* (bulk water resolved into a free layer, minutes at high field) and the *optical clock* (95% relative transmittance, slower by a factor 20–50), the latter controlled by a fines tail that no single-size argument can represent. Third, a dimensionless organizing group: the electrocoalescence number Λ, the induced-dipole pair energy at contact in units of k_BT. Λ is the droplet analogue of the dipolar coupling constant of electrorheology [7]; we show that it gates feasibility with a threshold predictable from shelf life, and that the published voltage-sweep data of Ref. [4] collapse onto $t \cdot \Lambda = \text{const}$ to within $\pm 5\%$. Fourth, a validation protocol we believe is itself useful: the prediction was made without calibration to any kinetic measurement, by three population-integration machineries with disjoint numerical failure modes (analytic, population balance, and stochastic simulation driven by a reinforcement-learning agent), with the measured kinetics quarantined until all three numbers were frozen. We are explicit below about what this protocol does and does not establish.

2 Results

2.1 A framework built from setup facts alone

We consider the batch configuration of Ref. [4]: 60 ml of water-in-hexadecane emulsion (0.5 M brine, 1% Span 80, water cut $\phi = 2\%$, DLS mean diameter 175 nm, lognormal width $\sigma_g = 1.35$) in a cubic cell of depth $H = 3.37$ cm, under a corona air-gap field of up to $E = 7.5$ kV/cm in the oil. Material properties (viscosity μ, densities, permittivity ε_{oil}, interfacial tension) are standard literature values (Methods). Crucially, no measured demulsification kinetics enter the model at any point; an older reduced-order model calibrated to the measured timescale was deliberately quarantined.

One further setup fact does quantitative work. The emulsion shows no visible phase separation for months without a field. Since the zero-field Brownian encounter rate is known, stability over a shelf life t_s bounds the per-encounter coalescence probability, and hence the surfactant film barrier: $\Delta G \gtrsim k_BT \ln(K_B n_0 t_s) \approx 19\, k_BT$ (instantiating "several months" as three; the criterion, fewer than one coalescence per drop, is stricter than "no visible separation" and thus a conservative lower bound). We carry $\Delta G = 20$–$30\, k_BT$ as the honest uncertainty band; Span-type monolayers stiffened by electrolytes sit in this range [8].

The pair physics is assembled from established elements: conductor-limit induced dipoles [6], Brownian transport with dipolar-capture enhancement (a Fuchs-type factor $\sim \Lambda^{1/3}$) [9, 10], near-contact amplification of the conductor–conductor attraction over the point-dipole law [11, 5], film-drainage barrier crossing, and, once drops grow past a micron, sweep coagulation: settling drops collect smaller partners from a capture tube whose radius is bounded jointly by energy ($U(r) = k_BT$) and by drift-versus-transit kinetics (Methods) [12, 10]. Merges of conducting drops in strong fields can also abort, ejecting charged daughter droplets; this partial-coalescence channel is gated by the electric Bond number at contact and by charge relaxation in the drop [13, 14, 15], and is retained because it carries the salinity dependence discussed below.

2.2 One number gates the cascade

Every stage of the chain is a contest between the induced-dipole bond and thermal noise. Dividing the aligned-pair contact energy by k_BT defines the electrocoalescence number,

$$\Lambda \;=\; \frac{\pi\, \varepsilon_0\, \varepsilon_{\text{oil}}\, K^2 E^2 a^3}{k_BT}, \tag{1}$$

with E the field in the oil, a the droplet radius, and $K \to 1$ the Clausius–Mossotti factor in the conductor limit. Λ is the droplet analogue of the dipolar coupling constant governing chain formation in electrorheological fluids [7]. Coalescence, however, must also defeat the film barrier. Two amplifications act at contact: a large drop presses a small one with $\sim 8\times$ the equal-pair energy (the "scavenging rule", Methods), and the near-contact conductor interaction exceeds the point-dipole law by a factor $A_{nc} \approx 2$–8 [11]. The go/no-go threshold is therefore

$$\Lambda^* \;\approx\; \frac{\Delta G / k_BT}{8\, A_{nc}} \;\approx\; 0.8 \qquad \text{(band 0.3–2)}, \tag{2}$$

predicted before any kinetic data are consulted. The two amplifications are not strictly independent (both are near-contact corrections to the point-dipole law), so their product should be read as a heuristic 16–64$\times$ band whose uncertainty the threshold inherits; ignition at $\Lambda \sim \Lambda^*$ additionally relies on the large-size tail of the initial distribution, since an equal median pair at threshold still faces a substantial residual barrier. Since $\Lambda \propto E^2 a^3$, the critical field steepens as droplets shrink, $E_{\text{crit}} \propto a^{-3/2}$ (Fig. 2): the compact statement of why nanoemulsions defeat conventional electrocoalescers and why the high-field air-gap architecture succeeds.

2.3 Calibration-free prediction, three ways

To make the prediction falsifiable we froze all model choices before consulting the measured kinetics, and solved the same physics by three machineries with disjoint numerical failure modes (Fig. 3). We stress that this is a no-calibration rather than a blind protocol: one author (S.P.) performed the source experiments and knew the measured clearing time. The safeguards are procedural, namely that no measured kinetic quantity enters the model at any point, that an older reduced-order model calibrated to the measured timescale was quarantined, and that all model choices and the three predicted numbers were fixed, under version control, before any comparison to data. The result therefore demonstrates that the framework requires no kinetic calibration, not that the modellers were ignorant of the answer; a hash-committed, pre-experiment version of this protocol would be stronger and we recommend it for future systems.

Route 1, analytic cascade. Closed-form staging: a Brownian gate ($\Lambda = 5.2$ at the 87.5 nm median radius and

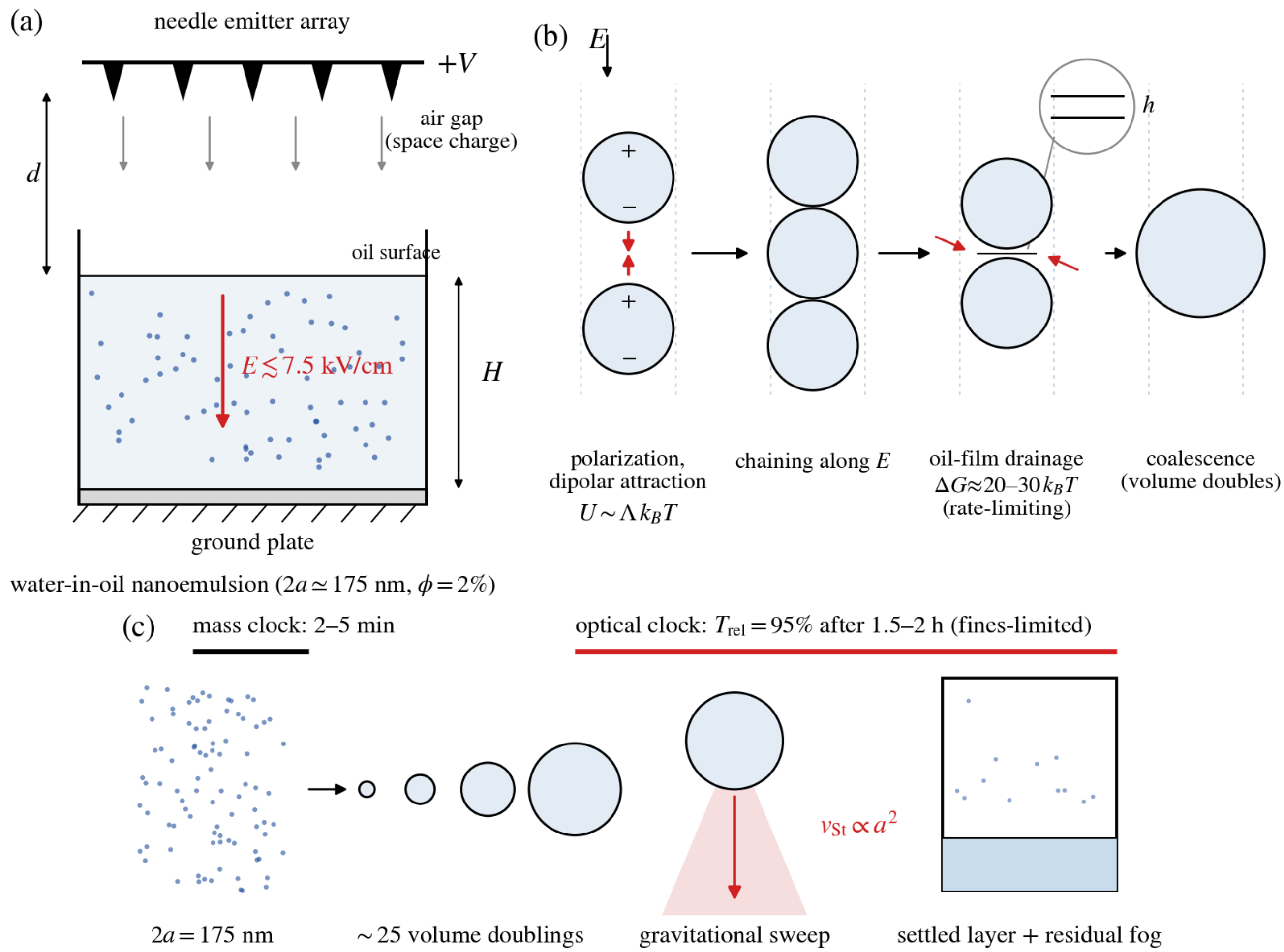

Figure 1: Setup and the multiscale chain from pair electrostatics to optical clearing. (a) The non-Laplacian batch configuration of Ref. [4]: a needle emitter array at $+V$ injects space charge across an air gap of width d onto a water-in-oil nanoemulsion of depth H above a grounded plate, sustaining fields up to $E \approx 7.5\,\mathrm{kV/cm}$ in the oil without shorting. (b) Pair-scale electrocoalescence: droplets polarize into induced dipoles, attract and chain along the field, drain the intervening oil film against the surfactant barrier $\Delta G \approx 20$–$30\,k_BT$ (the rate-limiting step), and merge. (c) Population scale: a parallel cascade of $\sim$25 volume doublings, with gravitational sweep dominating above a micron, delivers drops to Stokes settling; two clocks result, with bulk separation in minutes and optical clearing delayed by a sub-100-nm residual fog. The electrocoalescence number Λ gates the entire chain.

7.5 kV/cm: open; $\Lambda = 7.8$ at the 100 nm mean radius used for the data collapse), barrier-limited ignition (seconds), a growth ladder summing $\sim$25 volume-doubling times with Brownian-capture and sweep transport, and a gravity endgame at the handoff radius minimizing total time. Being single-sized, this route can only time the *mass* clock: 5 min (brine, barrier 20–25 k_BT), rising to $\sim$40 min at the 30 k_BT barrier edge.

Route 2, population balance. A 123-bin sectional Smoluchowski model [48, 49] with coagulation, Stokes settling loss, a daughter-droplet source from partial coalescence, and the endpoint evaluated exactly as defined operationally: relative transmittance from full Mie scattering [50] reaching 95% over a 1 cm path. Numerically, merges between comparable sizes are discrete fixed-pivot events while accretion of much smaller partners is treated as continuous growth (upwind advection in size space), which removes an otherwise fatal stiffness; mass is conserved to 1%.

Route 3, stochastic simulation with a learning agent. An independent super-droplet Monte Carlo implementation [52] exposes only operator-visible observables (turbidity, coarse mean size, free-water-layer height) to a tabular reinforcement-learning agent [53] that selects the field level every 30 s under an arcing risk above 8 kV/cm. With reward equal to negative elapsed time, the learned value of the initial state *is* the predicted clearing time. The agent independently rediscovers the experimental operat-

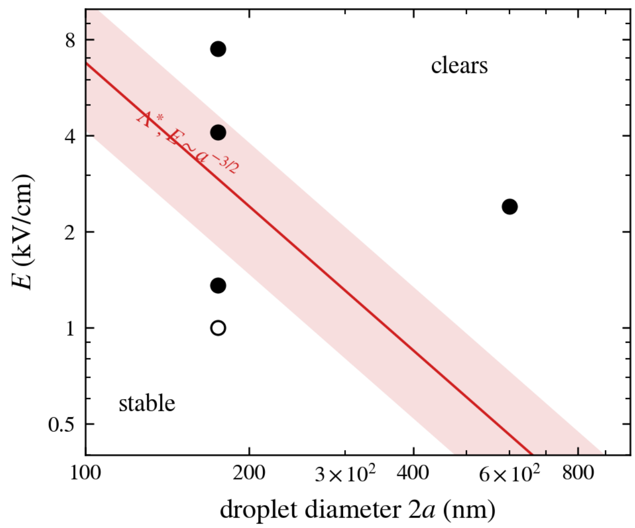

Figure 2: The feasibility map. The Λ^* boundary ($E \sim a^{-3/2}$; orange band spans $\Lambda^* = 0.3$–2) in field-versus-size space, with operating points from Ref. [4]: filled, cleared; open, remained milky. The sweep's cleared points bracket the boundary band, placing the operational threshold at the band's lower edge; a 600 nm emulsion clears at modest fields because $\Lambda \propto a^3$.

ing point, running at the arcing edge.

The three routes agree with one another within 21% (range over mean of the central estimates) on the optical clock: population balance 110 min for brine and 90 min for DI water (film-barrier band widens the brine value to 78 min–3.3 h), fixed-field super-droplet Monte Carlo 102 min (median over 20 random seeds, observed range 81–273 min), and learned value estimate 125 min. Throughout, the quoted prediction band 90–125 min is the range of central estimates across routes and water phases; it excludes the film-barrier systematic, which is reported separately. Cross-route disagreements during development were diagnostic rather than cosmetic: they exposed a mass leak at the size-grid boundary, an event-rate undercount in rare sweep interactions, and the stiffness pathology above, all of which a single-method study would have shipped silently. A thirteen-point verification suite (dimensional checks against raw constants, optical limits, grid-convergence, collapse statistics, and an AST-based solver-independence audit) accompanies the code and returns a nonzero exit status on any failure.

2.4 Two clocks, one process

The framework's central structural prediction is that "separated" is two different claims (Fig. ??). The cascade is parallel: all pairs merge simultaneously, the population halves per generation, and only $\log_2(4\times10^7)\approx 25$ generations connect 175 nm to settling sizes. Consequently bulk separation is fast: in the population balance, 97% of the water reaches the free layer within $\sim$2 min at 7.5 kV/cm (the analytic route's central estimate is 5 min). The transmittance endpoint, the standard optical route to emulsion stability assessment [46, 47], is another matter. A residual fog holding under half a percent of the water, droplets too small for even the amplified contact energy to open the film barrier, scatters strongly enough to keep the sample milky, and it can no longer coalesce in parallel: each straggler waits alone for a rare collector. Logarithmic physics for the bulk; linear waiting for the tail. The two clocks differ by a factor of 20–50, which is precisely why clearing times have resisted single-droplet reasoning: the eye's endpoint is a statement about the smallest decile of the initial size distribution, not about the bulk. Staged kinetics with a slow residual-emulsion tail have been reported in population-balance studies of microscale electrocoalescers [16]; the present contribution is to make the final stage quantitative through the scattering optics of the fines and to show it is the operational endpoint's rate-limiting step.

2.5 The data collapse onto one line

Reference [4] reports batch demulsification times over a threefold voltage sweep, fitted there as $t\sim V^{-1.91}$. At fixed droplet size the collapse below is therefore a dimensionless *restatement* of that fit rather than an independent measurement; its new content is the value and tightness of the constant, the a-priori threshold it terminates on, and the field-reading discrimination it enables. Digitizing the seven published points and converting voltage to Λ (field read as V/d; mean droplet radius $a = 100$ nm per the source's caption; electrode gap $d = 5.5$ cm per its Methods) yields

$$t_{95}\cdot\Lambda \;=\; 23.5\ \text{min}\quad(\text{RMS scatter }5\%,\ \text{range }21.5\text{–}25.5) \tag{3}$$

across the sweep (Fig. 6). Equivalently, the digitized exponent is $t \propto V^{-1.96\pm0.06}$, statistically indistinguishable from the E^{-2} scaling that Λ requires. Three qualifications bound this result. Because only V varies across the sweep, the collapse constrains the field exponent of Eq. 1 and nothing else; the a^3 and material-property content is untested by these data. The constant carries a systematic uncertainty of a factor $\approx$2 beyond its 5% scatter, since $C \propto a^3/d^2$ and depends on the choice of size-distribution moment. And the scatter itself is consistent with digitization noise alone (a Monte Carlo with 5% reading error in t and 2% in V reproduces it), so it should be read as an upper bound on the law's intrinsic scatter. Three further points follow. (i) The lowest voltage that still cleared sits at $\Lambda \approx 0.26$, and three of the seven cleared points lie below the central $\Lambda^* = 0.8$: the data are consistent with the a-priori threshold *band* of Eq. 2 only at its lower edge, indicating that the central near-contact amplification is too conservative and the true amplification is nearer the top of its 2–8 range. Below corona onset, where the field in the oil is negligible and Λ is effectively zero, no clearing was observed in 110 min, consistent with the gate. (ii) The collapse arbitrates a genuine ambiguity

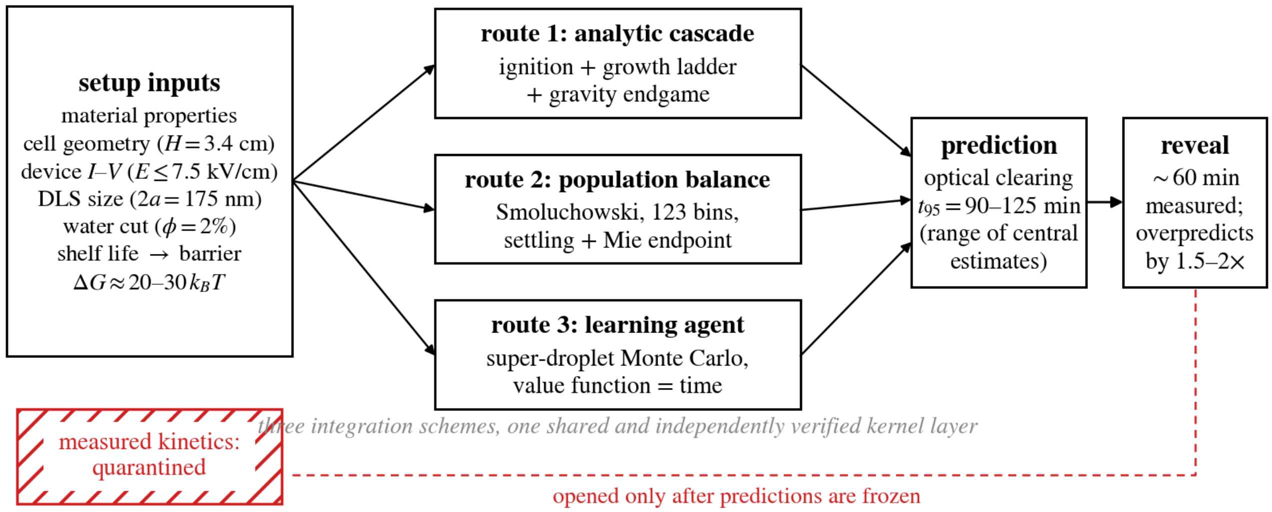

Figure 3: The blind-prediction protocol. Setup facts (materials, geometry, device characterization, size distribution, and a film-barrier bound from zero-field shelf life) feed three independent routes: a closed-form cascade, a population balance with Mie optics, and a super-droplet Monte Carlo rig driven by a reinforcement-learning agent. Measured kinetics remain quarantined until all three predictions are frozen.

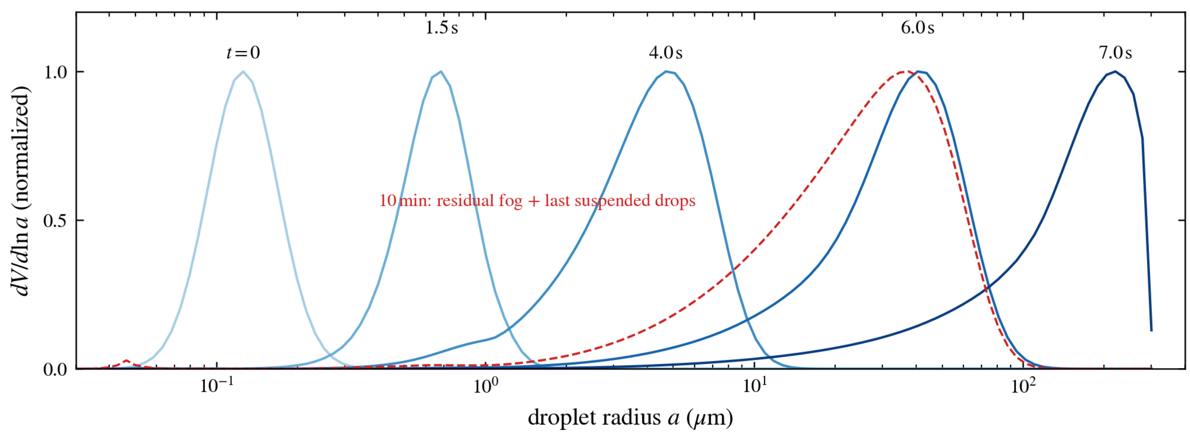

Figure 4: The coalescence cascade (population balance, 7.5 kV/cm, brine). Volume-weighted size distributions march through ~25 volume doublings within seconds (blue curves, labelled by time); by ten minutes (red dashed) the largest drops have settled out and the suspended remainder is the residual fog plus the last settling drops.

in the device physics. Writing the field as $(V - V_0)/d$ and profiling the collapse quality over the onset offset V_0 gives RMS spreads of 5.2, 8.9, 15.6, 33, and 58% at $V_0 = 0$, 1, 2, 4, and 6.3 kV: the data prefer $V_0 = 0$ and exclude offsets beyond ~1 kV at the level of the observed scatter, consistent with V/d approximating the average field in the oil, as the device's current–voltage characterization also indicates (the offset test varies V_0 only; it does not by itself pin the gap). (The corona *onset* of the source device remains physically real; what the data reject is a large standing offset in the oil-field reading above onset.) (iii) The collapse does *not* extend to the water-cut sweep of Ref. [4], nor should it: Λ answers whether a pair can merge, not how much water must move or how much light must clear; the water-cut axis requires a second, load-like group. Two further disclosures are essential. The collapse validates the E^{-2} scaling but *not* the framework's absolute kinetics across the sweep: at the sweep's low-field points our barrier-gated model predicts clearing times that are orders of magnitude too long (the exponential gate $e^{A_{nc}\Lambda - \Delta G/k_B T}$ effectively closes below ~3 kV/cm at this droplet size), whereas the experiments cleared in 11–95 min. The model's absolute kinetics are supported only near the high-field reference condition; the sweep data falsify the central barrier calibration at low fields, again pointing to stronger near-contact amplification or to field-

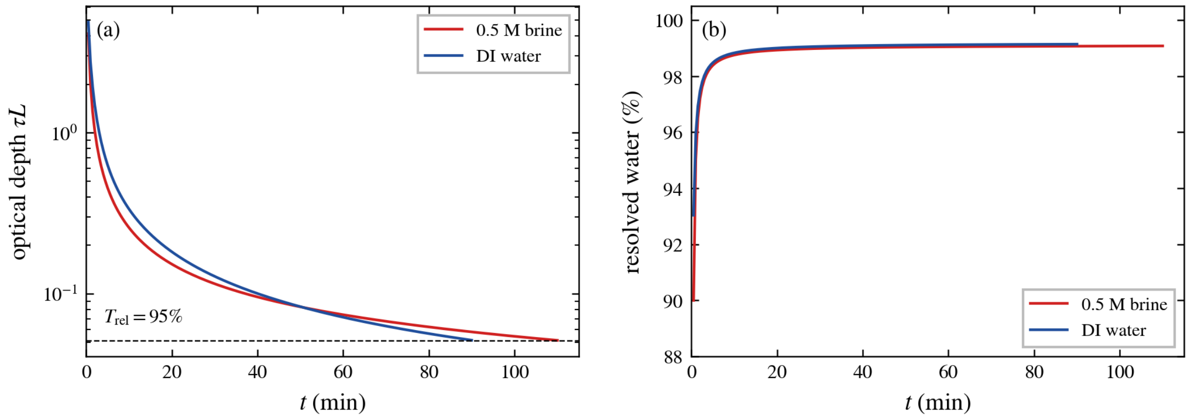

Figure 5: Two clocks (population balance, 7.5 kV/cm). (a) The optical clock: droplet optical depth decays toward the 95%-transmittance threshold (dashed) in 90–110 min, brine (red) slower than DI water (blue). (b) The mass clock: resolved water exceeds 97% within minutes for both phases.

driven transport (electrohydrodynamic circulation) that bypasses the gate. And the collapse is licensed only within the sweep's range: extrapolating $t = 23.5\,\mathrm{min}/\Lambda$ to the reference batch condition predicts minutes against the $\sim$60 min measured there, indicating that the sweep and the reference batch differ in effective field or geometry in a way the published information does not fully constrain. Resolving these tensions requires instrumented-field source data, and we flag them as the sharpest open questions the dimensionless analysis exposes.

On the absolute number the model *overpredicts*: the central band of 90–125 min lies a factor 1.5–2.1 above the single measured value of $\sim$60 min, with no stated experimental uncertainty or replicates to compare against, so the comparison establishes order-of-magnitude correctness with a systematic slow bias, not quantitative validation. Two mitigating observations bound its meaning. The omitted transport channels (electrohydrodynamic stirring, documented directly by the source experiments) act in the observed direction. And the endpoint itself is stiff: moving the transmittance threshold by one percentage point moves the predicted t_{95} by $-27\%/+46\%$ (81, 110, and 161 min at 94, 95, and 96%), so part of any gap at this endpoint reflects the endpoint's own sensitivity rather than cascade physics.

2.6 Why brine is the stubborn phase

A bench observation supplied as direction only, never fitted, is that saline droplets are markedly more stable under the field than fresh water. The framework was given the two literature mechanisms and the slowdown emerged in every route. First, electrolytes stiffen Span-type interfacial films: salting-out packs the sorbitan monolayer and raises its mechanical strength [8], the same interfacial-rigidity route by which asphaltene films stabilize crude emulsions [36], entering here as a few additional k_BT of barrier whose effect is exponential and therefore mild at 7.5 kV/cm but decisive at low field. Second, brine's conductivity ($\sim$4.6 S/m) allows charge to transfer within the coalescence bridge, so that high-field merges of grown drops abort into daughter droplets, the field-driven analogue of classical partial coalescence [33, 34, 35, 13, 14, 15]; fresh water relaxes too slowly to bounce. The net prediction, brine slower by $\sim$20% at 7.5 kV/cm (110 vs 90 min) with the penalty concentrated in the fines tail and growing toward lower fields, is testable: the film mechanism predicts a smooth, roughly exponential slowdown with ionic strength, while the bounce mechanism concentrates its damage at high field and late times, and each can be isolated by a salinity sweep at fixed field. We note that a naive electrostatic expectation (saltier drops polarize more strongly) predicts the opposite sign and is contradicted by the observation: above a conductivity of order $10\,\sigma_{\rm oil}$, the dipole is geometry-limited and salinity adds nothing to the driving force. Classical double-layer screening [38, 39, 37] is likewise secondary here, since the draining film is oil rather than electrolyte.

3 Discussion

The results support a simple claim with practical consequences: the clearing time of an electrostatically demulsified nanoemulsion is predictable from setup information alone, to engineering accuracy, and is organized by one dimensionless group. The two-clocks decomposition explains a long-standing frustration, namely why single-droplet intuition fails to anticipate macroscopic clearing: the operational endpoint is governed by the fines tail, an emergent population property. The Λ framework converts device design questions into geometry on the feasibility map: droplet size sets the required field through $E_{\rm crit} \propto a^{-3/2}$, and the measured collapse $t \approx 23.5\,\mathrm{min}/\Lambda$ prices the speed of any operating point for this emulsion class.

Three limitations bound the claims. Validation rests on one experimental system (seven voltage points, one

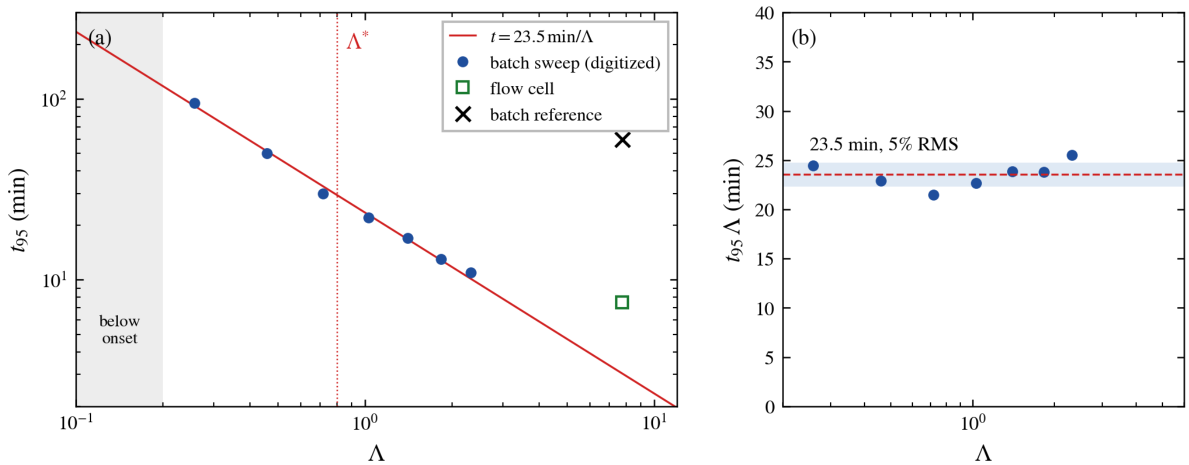

Figure 6: Validation against published data. Left: the seven digitized voltage-sweep points of Ref. [4] against $t = 23.5\,\mathrm{min}/\Lambda$; the flow-cell demonstration (open green) lies a factor 2.5 above the extrapolated line, and the batch reference measurement (orange cross) and the predicted band at the same Λ lie an order of magnitude above it, the extrapolation tension discussed in the text. Grey band: below corona onset, no clearing observed; dotted line: the a-priori threshold Λ^*. Right: the collapse, $t \cdot \Lambda$ with 5% RMS scatter.

absolute batch time, one flow-cell time); the near-contact amplification A_{nc} is an order-one parameter carried as a band rather than computed, and it sets the precise location of the feasibility cliff; and convective transport (electrohydrodynamic circulation, shear in flow cells) is omitted, plausibly explaining both the factor-of-two slow bias and the flow cell's additional speed. Each limitation is also a concrete measurement: dynamic light scattering on the residual haze after clearing would locate the fines cliff and hence A_{nc}; a salinity sweep at fixed field would separate the two brine mechanisms; and the learning agent's discovery that, under strong partial coalescence, a field *schedule* (ignite hard, then throttle to stop bouncing) beats any constant field is directly testable with a programmable supply.

A further caveat: the three routes share the same pair-physics kernel layer, so they are three population-integration schemes over a common, separately verified closure set; their mutual agreement (21%) guards against discretization and implementation error, which is exactly what the development history showed it catching, and not against a shared wrong model. Only the comparison with measurement tests the physics. More broadly, the no-calibration, triangulated protocol used here, three solvers with disjoint failure modes, predictions frozen before data contact, and a public verification suite, is inexpensive discipline for soft-matter kinetics modeling generally, where post-hoc calibration is endemic and plausible-but-wrong models are cheap to produce.

4 Materials and Methods

4.1 Setup parameters

All inputs are setup facts from Ref. [4] or literature property values: hexadecane $\mu = 3.03\,\mathrm{mPa\,s}$, $\rho_o = 773\,\mathrm{kg/m^3}$, $\varepsilon_{\mathrm{oil}} = 2.05$, $n_o = 1.434$; 0.5 M brine $\rho_w = 1017\,\mathrm{kg/m^3}$, $n_w = 1.338$, $\sigma_w = 4.6\,\mathrm{S/m}$; interfacial tension with Span 80, $\gamma = 4\,\mathrm{mN/m}$; oil conductivity $2 \times 10^{-8}\,\mathrm{S/m}$; cell depth $H = 3.37\,\mathrm{cm}$; optical path $L = 1\,\mathrm{cm}$; $T = 298\,\mathrm{K}$. Initial condition: lognormal, median diameter 175 nm, $\sigma_g = 1.35$, $\phi = 0.02$. The field is delivered by unipolar corona space-charge injection across the air gap [45, 4]. Drops are treated as spherical and non-deforming, valid since the operating field is well below the Taylor instability limit for these sizes [30, 31, 32]. Shear-driven collision kernels [43] and hindered-settling corrections [44] are neglected, consistent with a quiescent batch at $\phi = 2\%$; both would matter in flow systems or at high water cut.

4.2 Pair kernels

Aligned-pair contact energy $U_{ij} = 8\pi\varepsilon_0\varepsilon_{\mathrm{oil}}K^2E^2a_i^3a_j^3/(a_i + a_j)^3$, reducing to Λk_BT for equal spheres (Eq. 1); for $a_j \gg a_i$, $U \to 8\Lambda_i k_BT$ (the scavenging rule). Transport: Brownian kernel $K_B = (2k_BT/3\mu)(a_i + a_j)(a_i^{-1} + a_j^{-1})$ enhanced by the capture factor $\max(1, 1.12\,\Lambda_{ij}^{1/3})$ [9], plus sweep coagulation $K_g = \pi R_c^2|\Delta v_{\mathrm{St}}|$ with capture radius $R_c = \min\big(\Lambda_{ij}^{1/3}(a_i+a_j),\, (10A/\Delta v)^{1/4}\big)$, where $A = 4\varepsilon_0\varepsilon_{\mathrm{oil}}K^2E^2a_i^3a_j^3/(\mu\, a_{\min})$ is the dipole-drift coefficient; the second bound enforces that a partner must drift in within the transit time [12]. Coalescence probability per encounter: $\min\{1, \exp(A_{nc}\Lambda_{ij} - \Delta G/k_BT)\}$ with $A_{nc} = 4$ central (2–8 band) [11]; sub-unity coalescence efficiencies of this kind are standard in film-drainagelimited collision modeling [54, 40, 41, 33]. Partial coalescence: probability $g_\sigma \,\mathrm{Bo}^2/(\mathrm{Bo}^2 + \mathrm{Bo}_c^2)$ with electric Bond number $\mathrm{Bo} = \varepsilon_0\varepsilon_{\mathrm{oil}}(3E)^2 a_{\min}/\gamma$ and charge-supply gate $g_\sigma = \tau_b/(\tau_b + \tau_e)$. This functional form, the critical value $\mathrm{Bo}_c = 0.1$, the local-field factor 3, the $+4\,k_BT$ brine barrier increment, and the daughter recycling rule (half the smaller drop's volume at half its radius) are our model constructions, motivated by but not derived from Refs. [14, 13, 15, 8]; they carry order-one uncertainty and are exercised in the sensitivity bands. Settling: Stokes, $v_{\mathrm{St}} = 2\Delta\rho g a^2/9\mu$; removal rate v/H.

4.3 Route 1: analytic cascade

Ignition frequency from the lognormal-averaged kernel; growth as an explicit octave ladder, $t_{\mathrm{growth}} = \sum_k \tau_{\mathrm{oct}}(a_k)$ with $\tau_{\mathrm{oct}} = 2/(K_{\mathrm{eff}}n)$ and $n = 3\phi/4\pi a^3$; gravity endgame $t_{\mathrm{settle}} = (H/2)/v_{\mathrm{St}}(a_h)$ with handoff radius a_h minimizing the sum. The ladder is top-heavy in the Brownian regime and bottom-heavy in the sweep regime, so no single-rung approximation is valid.

4.4 Route 2: population balance

Sectional Smoluchowski equations [42] on 123 geometric bins (25 nm–300 μm), fixed-pivot deposition [49] for comparable-size merges; accretion (mass ratio $< 10^{-2}$) treated as continuous growth via upwind advection in size space with number conservation; overflow beyond the top bin routed to the resolved pool. Explicit conservative time stepping with $\Delta t \le 0.2/\max(\text{per-particle rate})$; the global mass audit closes at 0.99, and refining the grid from 123 to 246 bins shifts t_{95} by +3.8%, so discretization error is small against the reported bands (Fig. S1c). Endpoint stiffness is reported in Results and Fig. S1b; the barrier band and the onset-offset profile are shown in Figs. S1a and S1d. Endpoint: $T_{\mathrm{rel}} = \exp[-L\sum_i n_i C_{\mathrm{sca}}(a_i)]$ with Mie cross sections (relative index 0.933) [50].

4.5 Route 3: super-droplet Monte Carlo and learning agent

512 computational droplets with multiplicities; pair sampling with multiple-coalescence correction [52]; independent anomalous-diffraction optics [51]. The agent observes (turbidity bin, mean-size bin, resolved-layer quartile), selects among six field levels each 30 s (arc risk above 8 kV/cm), and is trained by every-visit Monte Carlo control with constant step size [53] (800 episodes, ϵ-greedy, undiscounted); in our experiments one-step Q-learning converged poorly here because the near-self-transition "field off" action remains spuriously competitive under bootstrapping. Predictions: the greedy-rollout distribution and $-\max_a Q(s_0, a)$, the latter carrying a maximization bias toward optimistic times that we note explicitly.

4.6 Verification and data

A twelve-check suite verifies dimensional arithmetic against raw constants, Mie limits (Rayleigh and geometric), Stokes values, initial opacity against the reported 0% transmittance, zero-field stability, the collapse statistics, the field-reading discrimination, and solver independence. The seven voltage-sweep points were digitized from Fig. 4C of Ref. [4] (gap 5.5 cm per its Methods; field read as V/d per its Fig. 4B). All code, parameters, and figure-generation scripts: `https://github.com/sreedath/emulsion-clock`.

Appendix: supplementary figures

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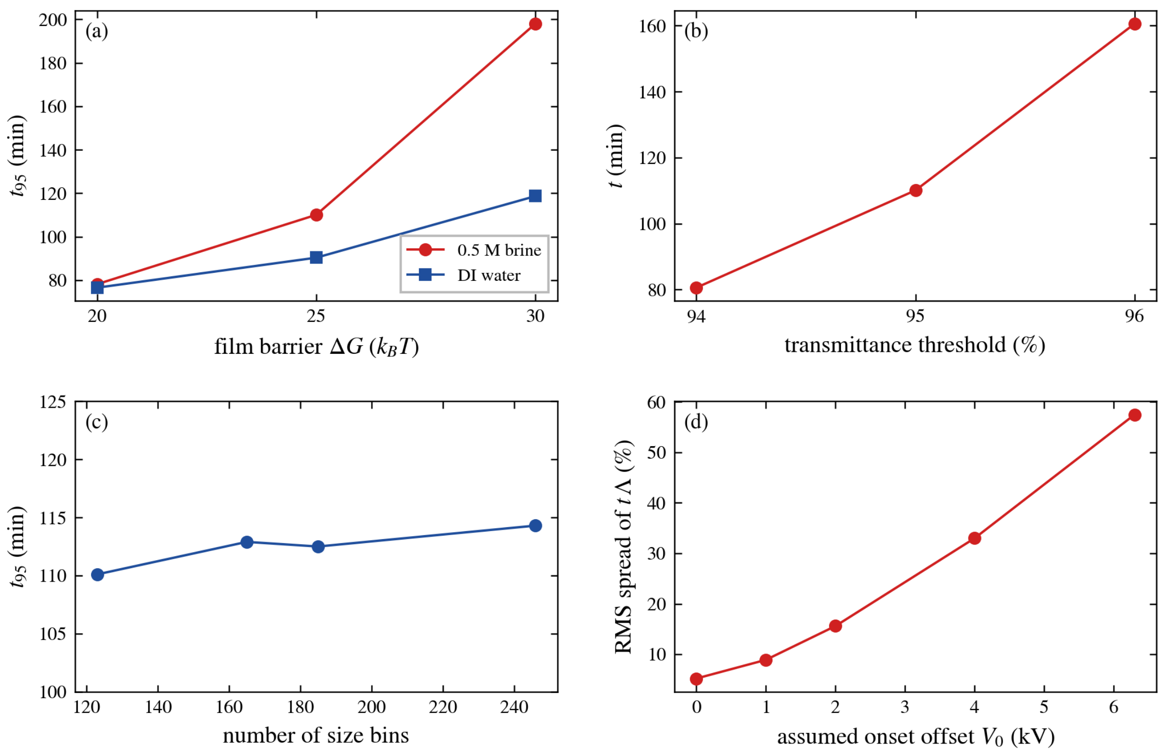

Figure S1: Robustness of the headline numbers. (a) Sensitivity of the predicted optical clearing time to the film barrier over its 20–30 $k_B T$ band, both water phases. (b) Stiffness of the operational endpoint: a one-percentage-point change in the transmittance threshold moves t_{95} by $-27\%/+46\%$. (c) Grid convergence of the population balance: doubling the size resolution shifts t_{95} by $+3.8\%$. (d) Onset-offset profile of the data collapse: the RMS spread of $t \cdot \Lambda$ across the digitized sweep degrades monotonically with an assumed corona offset V_0 in the field reading, preferring $V_0 = 0$.

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Data and code availability. All code, inputs, digitized data, and figure-generation scripts, under version control with the frozen prediction state (commit `ab555b0`), are available at https://github.com/sreedath/emulsion-clock. An interactive companion is available at https://demulsification-prediction.vercel.app.